MIL-C-20/1C
25 May 1973
SUPERSEDING
MIL-C-20/1B
11 April 1962

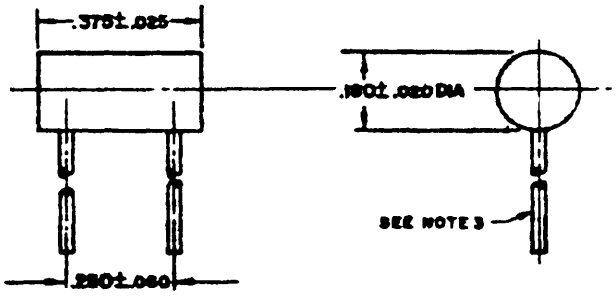
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (TEMPERATURE COMPENSATING) (UNINSULATED), NON-ESTABLISHED RELIABILITY, STYLE CC20

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-20.



INCHES	MM
.020	.51
.025	.64
.060	1.52
.180	4.57
.250	6.35
.375	9.53

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Leads shall be of tinned wire, .025±.002(.64±.05 mm) diameter and 1.250(31.75 mm) minimum length.

FIGURE 1. Style CC20 capacitors (uninsulated).

REQUIREMENTS:

Design and construction:
Dimensions and configuration: See figure 1.
Lead type: Radial.
DC rated voltage: 500 volts.
Operating temperature range: -55° to +85°C.
Characteristic: See table I.

Capacitance: Within tolerance specified (see table I).

Quality factor (Q): In accordance with MIL-C-20.

FSC 5910

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TABLE I. Capacitor characteristics.

Type designation 1/	Capacitance (pF)	Capacitance tolerance	Temperature coefficient shown as column head and tolerance envelope for temperature coefficient shown in column								
			A	C	H	L	P	R	S	T	U
CC20-K0R5C	0.5	C	K	K	K	K	K	K	K	K	K
CC20-KR75-	0.75	CD	K	K	K	K	K	K	K	K	K
CC20-K1R0-	1	CD	K	K	K	K	K	K	K	K	K
CC20-K1R5-	1.5	CD	K	K	K	K	K	K	K	K	K
CC20-K2R0-	2	CD	K	K	K	K	K	K	K	K	K
CC20-J3R0-	3	CD	J	J	J	J	J	J	J	J	J
CC20-4R0-	4	CD	H	H	H	H	H	H	H	J	J
CC20-5R0-	5	CDF	H	H	H	H	H	H	H	J	J
CC20-6R0-	6	CDF	H	H	H	H	H	H	H	J	J
CC20-7R0-	7	CDF	H	H	H	H	H	H	H	J	J
CC20-8R0-	8	CDF	H	H	H	H	H	H	H	J	J
CC20-9R0-	9	CDF	H	H	H	H	H	H	H	J	J
CC20-100-	10	CDFG	H	H	H	H	H	H	H	J	J
CC20-120-	12	GJK	H	H	H	H	H	H	H	J	J
CC20-150-	15	GJK	H	H	H	H	H	H	H	J	J
CC20-180-	18	GJK	H	H	H	H	H	H	H	J	J
CC20-200-	20	GJ	H	H	H	H	H	H	H	H	J
CC20-220-	22	GJK	H	H	H	H	H	H	H	H	J
CC20-240-	24	GJ	H	H	H	H	H	H	H	H	J
CC20-270-	27	FGJK	.....	.....	.....	H	H	H	H	H	J
CC20-300-	30	FGJ	.....	.....	.....	.....	H	H	H	H	J
CC20-330-	33	FGJK	.....	.....	.....	.....	.....	H	H	H	J
CC20-360-	36	FGJ	.....	.....	.....	.....	.....	H	H	H	J
CC20-390-	39	FGJK	.....	.....	.....	.....	.....	.....	H	H	J
CC20-430-	43	FGJ	.....	.....	.....	.....	.....	.....	.....	H	J
CC20UJ470-	47	FGJK	.....	.....	.....	.....	.....	.....	.....	.....	J
CC20UJ510-	51	FGJ	.....	.....	.....	.....	.....	.....	.....	.....	J
CC20UJ560-	56	FGJ	.....	.....	.....	.....	.....	.....	.....	.....	J
CC20UJ620-	62	FGJ	.....	.....	.....	.....	.....	.....	.....	.....	J

1/ Complete type designation will include additional letter symbols to indicate, where applicable, temperature coefficient, tolerance envelope for temperature coefficient, and capacitance tolerance.

Temperature coefficient and capacitance drift: In accordance with MIL-C-20.
For capacitors with values of 20 pF or less, the temperature-coefficient tolerance shall be determined in accordance with the curve shown on figure 2. For points below the curve, characteristics are not considered practicably measurable.

Moisture resistance: In accordance with MIL-C-20, except polarization is not applicable.

Life: In accordance with MIL-C-20, operating condition 1.

Marking:
Method I - In accordance with MIL-C-20, except for "JAN" brand. The beginning of the type designation shall be adjacent to the end having the inner-electrode terminal.
Method II - In accordance with MIL-C-20 and as shown on figure 3.

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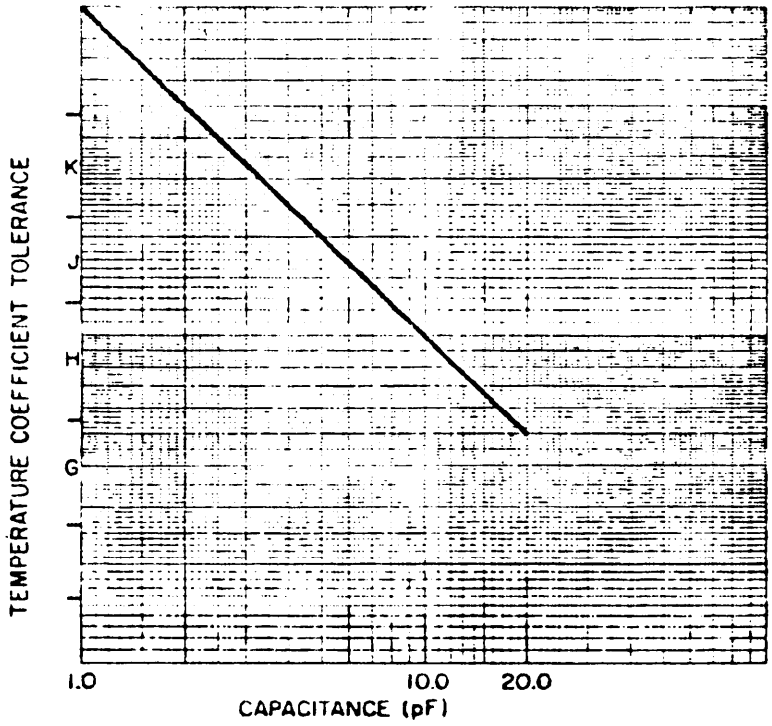
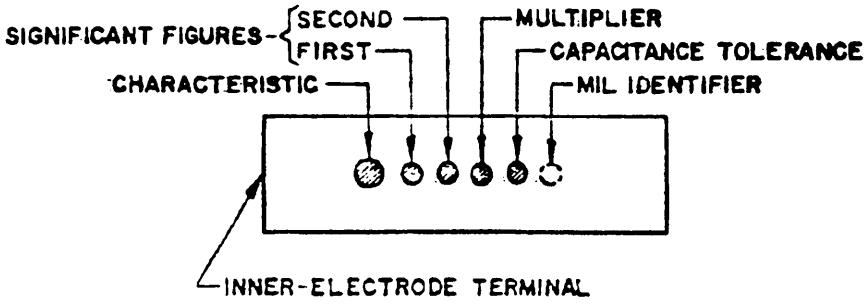


FIGURE 2. Limits for characteristic measurement.



1. The characteristic, capacitance, and capacitance tolerance shall be indicated on the capacitor by five spots of color. The characteristic spot shall be adjacent to the inner-electrode terminal; the diameter shall be at least twice that of any other spot, or the space from the characteristic spot to the adjacent end of the capacitor body shall be no larger than one-third the space from the capacitance-tolerance spot to the opposite end of the capacitor body.

2. The MIL identifier (black spot) shall be applied to the capacitor body diametrically opposite the color spots.

FIGURE 3. Color-code marking.

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Navy - EC
Air Force - 11

Review activities:
Armv -
Navy - OS
Air Force - 80
DSA - ES

User activities:
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Navy - AS. SH. MC
Air Force -

Preparing activity:
Air Force - 11

Agent:
DSA - ES

(Project 5910-1064-1)

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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
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b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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